

Consumer Shopping Behavior Analysis

Business Intelligence Performance Report

1. Executive Summary

This analysis evaluates customer purchasing behavior across **3,900 orders** to identify patterns in customer loyalty, product performance, demographics, promotional usage, customer satisfaction, shipping preferences, subscription behavior, and seasonality.

The analysis shows that the business is supported by a highly active repeat-customer base. Customers classified as **Loyal account for 3,116 of the 3,900 orders, approximately 80% of total order volume**. This indicates that repeat purchasing is a major component of overall transaction activity.

Product-category analysis shows that **Clothing is the largest category by order volume**, with 1,737 orders, while Clothing and Accessories were identified among the leading revenue-generating categories. The analysis also reveals meaningful differences in purchasing behavior across product categories, seasons, shipping methods, customer segments, and promotional usage.

Customer demographics are relatively balanced across the four age groups. However, the gender-level analysis revealed a notable promotional pattern: **all recorded discounted transactions in the dataset were associated with male customers, while no discounted transactions were recorded for female customers**. This is an important descriptive finding that warrants further investigation because the dataset alone cannot establish why this difference exists.

Promotional analysis further shows that discount usage is widespread, but its relationship with Average Order Value (AOV) varies by category. Footwear showed a positive AOV difference for discounted orders, while Clothing, Accessories, and Outerwear showed lower AOV among discounted orders.

Shipping behavior was relatively distributed across the available fulfillment options. **Free Shipping recorded the highest order volume and revenue**, with 675 orders generating \$40,777 in revenue and an AOV of \$60.41.

Overall, the analysis indicates that the business should move toward **behavior-based customer segmentation, category-specific commercial strategies, more targeted promotional activity, and continued monitoring of customer experience and fulfillment preferences**.

2. Business Problem

The business has access to detailed customer transaction data but requires a clearer understanding of how customers purchase, what products drive performance, which customer segments generate the greatest activity, and which commercial factors drive consumer decisions and repeat purchases.

The analysis was therefore designed to determine how the business can leverage consumer shopping data to identify trends, improve customer engagement, and optimize marketing and product strategies.

3. Data Overview & Methodology

Rows: 3900

Columns: 18

Key Features:

- Customer demographics (Age, Gender, Location, Subscription Status)

- Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
- Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type) - Missing Data: 37 values in Review Rating column

Analytical approach

The project began with data cleaning and exploration in Python. The dataset was imported using Pandas while `df.info()` and `df.describe()` were used to check the data structure and summary statistics.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	2	2
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	No	No
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	2223	2223
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN	NaN

- **Missing Data Handling:** Checked for null values and imputed missing values in the **Review Rating** column using the median rating of each product category.
- **Column Standardization:** Renamed columns to **snake case** for better readability and documentation.
- **Feature Engineering:**
 - Created **age_group** column by binning customer ages.
 - Created **purchase_frequency_days** column from purchase data.
- **Data Consistency Check:** Verified if **discount_applied** and **promo_code_used** were redundant; dropped **promo_code_used**.
- **Database Integration:** Connected Python script to MySQL and loaded the cleaned DataFrame into a database for SQL analysis.

MySQL was used for structured analysis and segmentation, while Power BI was used to transform the analysis into an interactive dashboard report.

The Power BI report was structured around two primary dashboard areas:

- Customer & Sales Overview
- Customer Behaviour

4. Core Performance Indicators

KPI	Result
Total Orders	3,900
Loyal Customer Orders	3,116
Loyal Share of Orders	~80%
Overall Average Order Value	~\$58–\$60 range
Overall Average Review Rating	3.76 / 5
Largest Category by Order Volume	Clothing

KPI	Result
Clothing Orders	1,737
Highest Shipping Order Volume	Free Shipping — 675 orders

These KPIs provide the baseline for interpreting the deeper customer and behavioral analysis.

5. Customer & Sales Performance

5.1 Overall Transaction Activity

The dataset contains **3,900 orders**, representing the complete transaction population available for this analysis.

The other useful measures are:

- Total Orders
- Revenue
- Average Order Value
- Product Category Performance
- Customer Segment
- Discount Behavior
- Customer Satisfaction

5.2 Product Category Performance

Product category analysis provides insight into where transaction activity is concentrated.

The four major categories in the dataset are:

- Clothing
- Accessories
- Footwear
- Outerwear

	category	total_orders	order_distribution_pct	Revenue	revenue_pct
►	Clothing	1737	44.54	104264	44.73
	Accessories	1240	31.79	31.79	31.83
	Footwear	599	15.36	36093	15.49
	Outerwear	324	8.31	18524	7.95
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Clothing

Clothing recorded **1,737 orders**, making it the largest category by transaction volume. This makes Clothing an important driver of overall sales activity and a key category for evaluating pricing, promotional, and customer-retention strategies.

Accessories

Accessories was also identified as one of the stronger revenue-generating categories. Because accessories are generally suitable for complementary purchases, the category presents an opportunity for cross-selling and bundle strategies.

Footwear

Footwear demonstrated an interesting relationship between promotions and transaction value. Discounted Footwear orders produced an AOV of **\$61.80**, compared with **\$59.08** for orders without discounts.

This represents an observed **+\$2.72 AOV difference**.

Outerwear

Outerwear produced the largest negative AOV difference associated with discount usage. Discounted orders averaged **\$55.32**, compared with **\$58.66** for non-discounted orders, representing an observed difference of **-\$3.34**.

5.3. Customer Segmentation & Loyalty

Customer segmentation was based on previous purchasing behavior:

- **New:** 1 previous purchase
- **Regular:** 2–10 previous purchases
- **Loyal:** More than 10 previous purchases

	category	total_orders	order_distribution_pct	Revenue	revenue_pct
▶	Clothing	1737	44.54	104264	44.73
	Accessories	1240	31.79	31.79	31.83
	Footwear	599	15.36	36093	15.49
	Outerwear	324	8.31	18524	7.95
	ON/SLM69L	354	8.37	18254	7.82

The most significant finding is the size of the Loyal customer segment.

Loyal customers accounted for:

3,116 of 3,900 orders = approximately 80% of total orders.

This demonstrates that repeat purchasing is a major feature of the customer base.

From a BI perspective, this means customer-retention behavior is particularly important when evaluating overall business performance.

The business should therefore monitor:

- Repeat purchase behavior
- Customer lifetime value
- Purchase frequency
- AOV by customer segment
- Category preference by segment
- Promotional dependence by segment

6. Demographic Analysis

6.1 Age Distribution

	age_group	num_of_orders	total_revenue
▶	Young Adult	1028	62143
	Middle Age	986	59197
	Adult	942	55978
	Senior	944	55763
	26U10L	844	22383

The customer population was distributed relatively evenly across the four age groups. No single age group dominates the dataset.

This suggests that the business serves a relatively broad demographic population rather than being concentrated within one particular age category.

6.2. Gender Analysis

	gender	total_orders	discounted_orders	discount_adoption_pct
▶	Male	2652	1677	63.24
	Female	1248	0	0.00

Gender analysis revealed an important behavioral and promotional pattern.

The customer base contains both male and female customers, but the promotional analysis produced a particularly notable finding:

All recorded discounted transactions were associated with male customers. No discounted transactions were recorded for female customers in the dataset.

The dataset does not provide sufficient information to determine whether this pattern resulted from:

- Promotional targeting
- Customer purchasing behavior
- Product selection
- Campaign exposure
- Data collection
- Another operational factor

This should be investigated before decisions are made about gender-based promotional strategy.

7. Discount & Promotional Analysis

Discount analysis was conducted to understand promotional adoption and whether discounted orders were associated with differences in AOV.

	age_group	total_orders	discounted_sales	discount_adoption_pct
▶	Senior	944	417	44.17
	Adult	942	406	43.10
	Young Adult	1028	443	43.09
	Middle Age	986	411	41.68

7.1 Discount Adoption by Customer Segment

Loyal customers recorded a **43.65% discount adoption rate**, compared with **40.43% among regular customers**.

The difference is relatively modest.

This suggests that although loyal customers make extensive use of promotions, their discount adoption is not dramatically different from that of other active customer groups.

7.2 Discount Adoption by Age

Discount adoption remained relatively consistent across age groups, ranging from approximately: **41.68% to 44.17%**.

The narrow range suggests that promotional usage is broadly distributed across the age groups. A formal statistical test would be required before concluding that age has no relationship with discount behavior. Descriptively, however, there is no large difference in adoption rates between the groups.

7.3 Discount Impact on AOV

The relationship between discount usage and AOV differs substantially across product categories.

	discount_applied	a_o_v	revenue	revenue_percentage	number_of_customers
►	Yes	59.2791	99411	62.96	1677
	No	59.9785	58479	37.04	975

Interpretation

Footwear is the only category where discounted orders show a higher AOV.

Clothing shows almost no difference.

Accessories and Outerwear show lower AOV among discounted orders.

These results indicate that **promotional effectiveness is category-dependent**.

The analysis does not establish that discounts caused the differences. It only identifies an association between discount status and observed transaction value.

8. Seasonality Analysis

Seasonality was examined to identify differences in customer purchasing behavior across seasonal periods and product combinations.

One notable result was **Fall Footwear**, which recorded an AOV of approximately: **\$63.71**

This represents one of the stronger category-season AOV combinations observed in the analysis.

Fall Footwear also recorded a discount usage rate of approximately **49.26%**, making it an interesting combination for further promotional analysis.

Rather than applying the same promotional strategy throughout the year, the business could use these seasonal patterns to identify product-season combinations where promotional activity and transaction value warrant further testing.

9. Shipping & Fulfillment Analysis

Shipping analysis examined order volume, revenue, and AOV across available fulfillment methods.

	shipping_type	average_purchase_amount	total_revenue	total_orders
►	Free Shipping	60.41	40777	675
	Standard	58.46	38233	654
	Store Pickup	59.89	38931	650
	Next Day Air	58.63	37993	648
	Express	60.48	39067	646
	2-Day Shipping	60.73	38080	627

Free Shipping recorded the highest order volume and highest total revenue among the shipping options.

It also maintained an AOV of **\$60.41**, demonstrating that the highest-volume shipping option was not associated with a substantially lower transaction value.

The relatively narrow AOV range across shipping methods suggests that customer spending is broadly similar regardless of fulfillment choice.

10. Customer Satisfaction & Reviews

	age_group	avg_rating
▶	Middle Age	3.71
	Young Adult	3.8
	Senior	3.75
	Adult	3.73
	all	3.75

Customer satisfaction was assessed using the available review-rating data.

The overall average review rating was: **3.75 / 5**

This provides a useful baseline for monitoring customer experience.

The rating metric should be considered alongside sales and behavioral measures rather than in isolation.

11. Customer Subscription / Membership Behavior

Subscription or membership status was included as part of the customer-behavior analysis.

This dimension is relevant because subscription status can provide another way of understanding customer engagement beyond purchase frequency alone.

The analysis should therefore be interpreted alongside:

- Customer segment
- Previous purchases
- Discount behavior
- AOV
- Product preferences
- Shipping preferences

Rather than treating subscription status as an isolated metric, it can be used as an additional behavioral attribute when developing customer profiles.

12. Cross-Analysis: *(What the Data Says Collectively)*

The strongest value of this project comes from combining the individual analyses rather than interpreting every variable independently.

Insight 1: The business has a strong repeat-purchase base

Approximately **80% of orders come from customers classified as Loyal**.

This makes retention and repeat purchasing important components of the business model.

Insight 2: Product performance is not uniform

Clothing generates the largest order volume, while category-level revenue and AOV patterns differ.

Therefore, commercial strategies should not necessarily be applied uniformly across all products.

Insight 3: Promotional effectiveness varies by category

Footwear shows a positive AOV difference associated with discounts, while Accessories and Outerwear show negative differences.

This supports a move toward **category-specific promotional analysis**.

Insight 4: Promotional exposure differs sharply by gender

The dataset contains discounted transactions associated with male customers but **none associated with female customers**.

This is one of the most important demographic findings and warrants further investigation.

Insight 5: Age groups behave relatively similarly

The four age groups are relatively balanced in size, and discount adoption varies only within a narrow range.

This suggests that age alone may not provide strong differentiation for promotional strategy.

Insight 6: Free Shipping is commercially significant

Free Shipping generated the highest order volume and revenue while maintaining an AOV of **\$60.41**.

This makes shipping strategy an important area for continued monitoring.

Insight 7: Customer satisfaction has room for improvement

An overall rating of **3.76/5** provides a useful baseline and should be monitored alongside category, fulfillment, and customer-segment performance.

13. Recommendations

13.1 Develop Behavior-Based Customer Segmentation

Because Loyal customers account for approximately 80% of orders, customer lifecycle and purchasing behavior should play an important role in segmentation.

The business should prioritize variables such as:

- Purchase frequency
- Previous purchases
- AOV
- Category preference
- Discount behavior
- Subscription status

rather than relying primarily on demographic characteristics.

13.2 Introduce Category-Specific Promotional Strategies

The analysis demonstrates that discount behavior differs across product categories.

Recommended direction:

Footwear: Continue testing targeted promotional campaigns because discounted orders currently show a higher observed AOV.

Clothing: Test whether promotional intensity can be reduced without negatively affecting order volume.

Accessories: Test cross-selling and bundle strategies.

Outerwear: Test bundles and threshold-based promotions rather than relying primarily on direct price reductions.

13.3 Test Threshold-Based Promotions

Rather than relying exclusively on unconditional discounts, the business could test offers such as:

“Get 10% off when you spend \$75 or more.”

With typical AOV concentrated around approximately \$58–\$60, a higher spending threshold could be tested to determine whether customers increase basket size.

This should be evaluated using:

- AOV
- Units per order
- Revenue per order
- Discount cost
- Gross margin
- Repeat purchase rate

An A/B test would provide stronger evidence than observational comparisons alone.

13.4 Investigate Gender-Based Promotional Differences

The absence of discounted transactions among female customers is a significant data observation. Before implementing a new gender-based promotional strategy, the business should investigate:

- Whether female customers were exposed to promotional campaigns
- Whether product/category mix explains the difference

- Whether campaign targeting created the disparity
- Whether the dataset accurately captures all promotional activity
- Whether the pattern exists consistently over time

The objective should be to understand the underlying business process rather than assume a cause from the transaction data alone.

13.5 Continue Monitoring Free Shipping

Free Shipping recorded:

- **675 orders**
- **\$40,777 revenue**
- **\$60.41 AOV**

Because it has the highest order volume and revenue among shipping methods, its performance should continue to be monitored.

Further analysis could determine whether free shipping is associated with higher repeat purchasing or customer satisfaction.

13.6 Use Customer Ratings as a Supporting Performance KPI

The overall **3.76/5 average rating** should be incorporated into ongoing customer-performance monitoring. Rather than treating ratings as a standalone customer-service metric, management should connect them with:

- Product category
- Shipping method
- Customer segment
- Season
- Discount status

This can help identify operational areas where customer experience may be affecting commercial performance.

14. BI KPI Framework for Future Monitoring

Business Area	KPI
Sales	Total Revenue
Sales	Total Orders
Sales	Average Order Value
Customer	Orders by Customer Segment
Customer	Repeat Purchase Rate
Customer	Customer Lifetime Value
Product	Revenue by Category
Product	AOV by Category
Promotions	Discount Adoption Rate
Promotions	Discounted vs Non-Discounted AOV
Fulfillment	Orders by Shipping Type
Fulfillment	Revenue by Shipping Type
Customer Experience	Average Review Rating
Customer Experience	Rating by Category
Loyalty	Subscription/Membership Rate
Seasonality	Revenue by Season
Seasonality	AOV by Season

15. Data Limitations

The findings should be interpreted within the limitations of the available dataset.

15.1 No Sales Channel Data

The dataset does not contain an online/offline sales-channel field. Channel performance therefore could not be analyzed.

15.2 No Product Cost or Margin Data

The analysis contains revenue and AOV information but does not contain product cost or gross-margin information.

Therefore, the analysis can identify differences in transaction value but cannot directly quantify profit impact.

15.3 One Order per Customer ID

Each customer ID occurs once in the dataset.

Consequently, the dataset represents each customer as a single transaction record, and customer count should not be interpreted as a separate measure from order count.

15.4 Observational Analysis

The analysis identifies patterns and associations in the transaction data.

It does not establish causality. For example, a lower AOV among discounted orders does not prove that the discount caused customers to spend less.

15.5 Promotional Exposure

The gender-level discount finding should be investigated further because the dataset shows a difference in observed promotional usage but does not explain the operational reason behind it.

15.6 Statistical Testing

Most findings in this project are descriptive BI findings. Where statements about statistical significance or predictive relationships are required, additional statistical testing should be performed.

16. Recommended Next-Level Analysis

To extend this project beyond descriptive BI reporting, the following analyses would provide additional business value:

- **Customer Lifetime Value**
Estimate the long-term economic value of New, Returning, and Loyal customers.
- **Cohort Analysis**
Track customer retention based on the period of first purchase.
- **Promotional A/B Testing**
Compare different discount structures to determine incremental impact on AOV and revenue.
- **Basket Analysis**
Identify products frequently purchased together and develop cross-selling opportunities.
- **Statistical Testing**
Test whether differences in discount usage, AOV, ratings, and demographic behavior are statistically significant.
- **Profitability Analysis**
Add product cost and margin data to determine whether promotional strategies actually improve profitability.

17. Conclusion

The Consumer Shopping Behavior analysis provides a broad view of how customers interact with the business across products, customer segments, demographics, promotions, fulfillment, satisfaction, and seasonality.

The most significant finding is the strength of the repeat-customer base: **Loyal customers account for approximately 80% of all orders.** This makes customer retention and repeat purchasing central components of the business's performance.

At the product level, **Clothing represents the largest transaction-volume category**, while category-level analysis reveals different AOV and promotional patterns across Clothing, Accessories, Footwear, and Outerwear.

The promotional analysis demonstrates that discounts do not have a uniform relationship with transaction value. Footwear shows a positive AOV difference, while Accessories and Outerwear show negative differences. This supports the use of more targeted promotional strategies rather than assuming that one approach will work equally across all categories.

The demographic analysis also identifies a notable difference in promotional exposure: **no discounted transactions were recorded for female customers in the dataset**, while discounted transactions were observed among male customers. This should be investigated further before drawing conclusions about the reason for the disparity.

Free Shipping also emerged as an important fulfillment option, recording the highest order volume and revenue while maintaining an AOV of \$60.41.

Overall, the analysis demonstrates the value of using Business Intelligence to move beyond basic sales reporting. By combining customer segmentation, product performance, demographic analysis, promotional behavior, fulfillment, customer satisfaction, and seasonality, the business can identify areas for deeper investigation and make more targeted, data-informed commercial decisions.

The next stage should be to connect these descriptive findings with **margin data, controlled promotional experiments, customer lifetime value, and statistical testing.** This would allow the business to move from identifying patterns to measuring the financial and operational impact of specific decisions.